

## SUDDEN DEATH - ASHCOMBE OBSERVATORY

|          |                         |              |                  |          |                        |
|----------|-------------------------|--------------|------------------|----------|------------------------|
| INCIDENT | 26-01-0117              | REPORTED     | 07:14 WED 07 JAN | OFFICER  | PS G. ORMEROD          |
| LOCATION | ASHCOMBE<br>OBSERVATORY | SUB-LOCATION | MAIN DOME        | DECEASED | HALE, Prof. E.<br>(62) |

## CIRCUMSTANCES

At 07:14 the caretaker, Thomas BRACKLEY, telephoned to report that the observatory's director, Professor Edmund HALE, was lying dead on the floor of the main telescope dome.

Mr Brackley states that he found the dome door bolted from the inside at about 07:05. He fetched a crowbar from the plant room and forced the door at 07:10. The bolt keeper was torn from the frame. He found Professor Hale at the foot of the observing platform ladder.

The dome has one door. The only other opening is the observing slit in the roof, which was open to the sky and is six metres above the floor. **The room was locked from the inside, with the deceased alone in it.**

## SCENE

The deceased lay on his back on the steel floor plate beside the base of the platform ladder. The platform is 2.4 m high and is reached by a vertical steel ladder. The dome is unheated; the scene thermometer read -3 C. The dome lights were off; his head torch was on, battery nearly flat. His laptop was open on the platform desk, screen asleep. His spectacles were found under the platform.

## OFFICER'S SUMMARY

Professor Hale was working alone in a locked dome at night, in the cold and the dark, on a ladder. Injuries are consistent with a fall from the platform (see B-1). No third party could have entered or left the dome. **Provisional finding: accidental death.** Referred to the coroner. No further police action.

**Reopened 02 MAR.** The deceased's sister has written to the coroner stating that her brother "had climbed that ladder ten thousand times." Referred to the Case Review Unit with the coroner's consent.

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POST-MORTEM EXAMINATION - HALE, EDMUND

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|      |              |          |               |          |              |
|------|--------------|----------|---------------|----------|--------------|
| CASE | OCME 26-0031 | DECEDENT | HALE, E. 62 M | EXAMINED | 08 JAN 10:20 |
|------|--------------|----------|---------------|----------|--------------|

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**INJURIES**

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**1. Occipital (back of the head).** Linear fracture of the occipital bone with an overlying laceration, 40 mm. Consistent with the back of the head striking a hard, straight edge, such as the edge of the steel floor plate.

**2. Vertex (crown of the head).** A circular depressed fracture, **34 mm across**, with a **fine criss-cross patterned abrasion** on the scalp over it. Attributed to the head striking a ladder fitting during the fall.

Either injury could have caused death. Bruising around both injuries shows that both were received while he was alive.

**OTHER FINDINGS**

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No defensive injuries. No injury to the hands or wrists. Blood alcohol nil. Toxicology negative. Mild coronary artery disease, not contributory.

**TIME OF DEATH**

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The body lay in an unheated dome at about -3 C with the roof slit open. Cooling in these conditions is too fast and too uneven for a useful estimate. **Death occurred between about 23:00 on 06 January and 03:00 on 07 January.** Any narrower time must come from other evidence.

**CONCLUSION**

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**Cause of death:** head injury. **Manner:** provisionally accidental, on the police account that he was alone in a room locked from inside.

**Note.** A person falling from 2.4 m would usually strike the head once. Two head injuries on opposite surfaces of the skull, one of them patterned, are unusual in a fall. I have accepted the police account of the scene.

T. BRACKLEY - CARETAKER

|             |                      |              |               |           |                          |
|-------------|----------------------|--------------|---------------|-----------|--------------------------|
| INTERVIEWEE | BRACKLEY, T.<br>(57) | DATE         | 07 JAN        | COMMENCED | 10:15                    |
| STATUS      | WITNESS              | INTERVIEWING | PS G. ORMEROD | ROLE      | CARETAKER, LIVES ON SITE |

- TRANSCRIPT BEGINS 10:15 -

**PS O** Take me through this morning.

**TB** I come up from the cottage at seven every day to check the boilers. The professor's coat was still on the hook, so I knew he'd stayed up there all night. I went to the dome to see if he wanted tea and the door wouldn't open.

**PS O** Wouldn't open how?

**TB** Bolted. There's a keypad lock and a hand bolt on the inside. I put my code in, the lock clicked, and the door still didn't move. That's the bolt. So I knocked, I shouted, nothing.

**PS O** And then?

**TB** I got the bar from the plant room and forced it. Ten past seven, about. Tore the keeper clean off. He was lying by the ladder. I rang you straight away.

**PS O** Was anyone else in the building overnight?

**TB** Dr Lindqvist was using the control room, the Swedish gentleman. He'd gone by the time I came up. The student went home before midnight.

**PS O** Did you speak to anyone last night?

**TB** No. I was in bed by eleven. Nobody rang me.

**PS O** You look after the door system?

**TB** I do. I'll print you off the dome door log. It'll show nobody went in after the professor.

- TRANSCRIPT ENDS 10:31 -

DR I. LINDQVIST - VISITING RESEARCHER

|             |                       |              |               |             |                             |
|-------------|-----------------------|--------------|---------------|-------------|-----------------------------|
| INTERVIEWEE | LINDQVIST, I.<br>(39) | DATE         | 07 JAN        | COMMENCED   | 14:40                       |
| STATUS      | WITNESS               | INTERVIEWING | PS G. ORMEROD | AFFILIATION | UPPSALA, VISITING<br>FELLOW |

- TRANSCRIPT BEGINS 14:40 -

- PS

O

You were here last night.
- IL

All night, in the control room. I run the telescope from the computer there. Edmund was up in the dome with the optics. He liked to tinker while I imaged.
- PS

O

When did you last see him?
- IL

About ten o'clock. He sat with me in the control room for an hour, then went up to the dome. After that I did not see him again.
- PS

O

Did you go up to the dome at any point?
- IL

No. Never. I was at my desk the whole time. My imaging sequence ran until about two in the morning and I watched every frame come in.
- PS

O

Did you leave the control room at all?
- IL

Once, at about quarter to one, I went out to my car by the side door for a phone charger. Two minutes. Otherwise no.
- PS

O

Did you hear anything from the dome?
- IL

Nothing. The dome motor is loud and the control room door was shut. I left at about three and drove to my lodgings.
- PS

O

How did you get on with Professor Hale?
- IL

Very well. He was a generous colleague. He was reviewing my comet paper for me.

- TRANSCRIPT ENDS 14:58 -

DR M. OKAFOR - DEPUTY DIRECTOR

|             |                           |              |               |           |        |
|-------------|---------------------------|--------------|---------------|-----------|--------|
| INTERVIEWEE | OKAFOR, M. (48)           | DATE         | 08 JAN        | COMMENCED | 09:05  |
| STATUS      | WITNESS - BY<br>TELEPHONE | INTERVIEWING | PS G. ORMEROD | LOCATED   | GENEVA |

- TRANSCRIPT BEGINS 09:05 -

**PS O** Where were you on the night of the sixth?

**MO** Geneva. I flew out on the fifth for a conference. I'm still here. I'll be back on Friday.

**PS O** You'll take over as director now.

**MO** Acting director. I know how that sounds. Edmund and I disagreed about plenty, but I would not have wished this on him.

**PS O** Did he have any worries recently?

**MO** He was troubled about something. He told me last week he'd found "a problem in somebody's data" and that he would have to deal with it before the journal did. He wouldn't say whose.

**PS O** Is the ladder dangerous?

**MO** Not for him. He wore a head torch and he never carried anything up it. Anything heavy went up on the hoist.

- TRANSCRIPT ENDS 09:19 -

E. HALE - WIFE OF THE DECEASED

|             |               |              |               |           |                    |
|-------------|---------------|--------------|---------------|-----------|--------------------|
| INTERVIEWEE | HALE, E. (59) | DATE         | 07 JAN        | COMMENCED | 16:30              |
| STATUS      | WITNESS       | INTERVIEWING | PS G. ORMEROD | ADDRESS   | HALLOW LANE, 19 KM |

- TRANSCRIPT BEGINS 16:30 -

**PS O** When did you last speak to your husband?

**EH** I rang him at quarter past twelve, on the dome phone. He sounded tired and pleased with himself. He said the seeing was the best all winter and he'd be home for breakfast.

**PS O** Did he mention anyone else?

**EH** He said, "I'm going to have a difficult conversation with Ivo before he goes." I didn't ask. He hated being asked.

**PS O** Where were you overnight?

**EH** At home, asleep. I'm on my own there when he's observing.

**PS O** There is a life insurance policy.

**EH** There is. Two hundred thousand. We took it out when we bought the house. You can look at whatever you like.

- TRANSCRIPT ENDS 16:44 -

K. NOWAK - DOCTORAL STUDENT

|             |                |              |               |            |               |
|-------------|----------------|--------------|---------------|------------|---------------|
| INTERVIEWEE | NOWAK, K. (26) | DATE         | 07 JAN        | COMMENCED  | 12:10         |
| STATUS      | WITNESS        | INTERVIEWING | PS G. ORMEROD | SUPERVISOR | PROF. E. HALE |

- TRANSCRIPT BEGINS 12:10 -

**PS O** You were at the observatory yesterday evening.

**KN** I had the telescope from eight until half eleven for my variable stars, then Dr Lindqvist took over. I left about twenty to twelve and got the last bus.

**PS O** How was Professor Hale?

**KN** Strange. He and Dr Lindqvist had words in the corridor at about ten. Quiet words, but sharp. Edmund said, "Tonight, Ivo. Not tomorrow." Then he went up to the dome.

**PS O** Your thesis. Professor Hale was going to fail it?

**KN** He wanted another year of data. That's not failing. I was upset, I told people I was upset. I wasn't going to hurt him over it.

**PS O** Did you come back to the observatory at any point?

**KN** No. I was home by ten past twelve.

- TRANSCRIPT ENDS 12:21 -

MAIN DOME DOOR - LOCK EVENT LOG

KeyLine K-40 lock controller, door MD-1. Printed from the site console, 07 JAN 09:40, by user T.BRACKLEY. View: **FILTERED**. Period 06 JAN 18:00 to 07 JAN 09:00.

| SEQ   | TIMESTAMP       | EVENT       | SOURCE                 |
|-------|-----------------|-------------|------------------------|
| #4468 | 06 JAN 18:00:00 | SELF-TEST   | CONTROLLER · OK        |
| #4469 | 06 JAN 18:00:01 | LATCH ARMED | CONTROLLER             |
| #4470 | 06 JAN 21:58:10 | OPEN        | KEYPAD · CODE H (HALE) |
| #4471 | 06 JAN 21:58:31 | CLOSED      | DOOR SENSOR            |
| #4478 | 07 JAN 07:31:05 | OPEN        | DOOR SENSOR · NO LATCH |

Event count for period: **11**. The controller numbers every event in order. It records every keypad entry, every opening from inside (handle sensor) and every closing. The hand bolt on the inside is mechanical and is not recorded.

CODES ISSUED

| CODE | HOLDER                                             |
|------|----------------------------------------------------|
| H    | Prof. E. HALE, director                            |
| O    | Dr M. OKAFOR, deputy director                      |
| B    | T. BRACKLEY, caretaker (also system administrator) |
| L    | Dr I. LINDQVIST, visiting (issued 02 JAN)          |
| N    | K. NOWAK, student                                  |

The controller also sends every event to the manufacturer's cloud service, which keeps its own copy. Site users cannot edit the cloud copy.

# TELESCOPE CONTROL LOG

Every command to the telescope mount is recorded with where it came from: the control-room software or the hand paddle on the platform in the dome.

| TIME     | SOURCE             | EVENT                                                 |
|----------|--------------------|-------------------------------------------------------|
| 20:02:15 | CONTROL ROOM       | Session K.NOWAK. Slew to target field V1342.          |
| 23:31:50 | CONTROL ROOM       | Session K.NOWAK ends.                                 |
| 23:36:04 | CONTROL ROOM       | Session I.LINDQVIST. Slew to target C/2025-K4.        |
| 23:38:40 | CONTROL ROOM       | Imaging sequence started, 60 frames.                  |
| 01:20:05 | CONTROL ROOM       | Imaging sequence complete, 60 of 60 frames.           |
| 01:31:40 | HAND PADDLE (DOME) | Manual slew east.                                     |
| 01:37:12 | HAND PADDLE (DOME) | Manual stop. Sidereal tracking resumed.               |
| 03:52:30 | MOUNT              | Horizon limit reached. Tracking halted automatically. |
| 07:44:10 | CONTROL ROOM       | Session POLICE. Log exported.                         |

No command of any kind was received between 01:37:12 and 03:52:30. A user at the telescope would normally stop the mount before it reaches the horizon limit.

MAIN BUILDING - CARD ACCESS LOG

Castlegate monitors the observatory's main entrance and its single side fire door. Both doors lock automatically. From outside, the main entrance opens only to a card; the side door does not open from outside at all. This system is run by Castlegate and cannot be changed by site staff.

| TIMESTAMP       | DOOR           | EVENT                              |
|-----------------|----------------|------------------------------------|
| 06 JAN 18:40:22 | MAIN ENTRANCE  | IN · E.HALE                        |
| 06 JAN 19:45:09 | MAIN ENTRANCE  | IN · K.NOWAK                       |
| 06 JAN 21:05:51 | MAIN ENTRANCE  | IN · I.LINDQVIST                   |
| 06 JAN 23:41:05 | MAIN ENTRANCE  | OUT · K.NOWAK                      |
| 07 JAN 00:46:58 | SIDE FIRE DOOR | OPENED FROM INSIDE · ALARM         |
| 07 JAN 00:49:12 | SIDE FIRE DOOR | CLOSED · ALARM RESET AUTOMATICALLY |
| 07 JAN 03:05:44 | MAIN ENTRANCE  | OUT · I.LINDQVIST                  |
| 07 JAN 07:02:30 | MAIN ENTRANCE  | IN · T.BRACKLEY                    |
| 07 JAN 07:29:40 | MAIN ENTRANCE  | IN · POLICE (VISITOR CARD)         |

No card belonging to M. OKAFOR or E. HALE (Mrs) was presented in this period. The side fire door can be propped open from inside; it was not propped (sensor closed throughout except 00:46 - 00:49).

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## SCENE & EXHIBITS - MAIN DOME

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### E-1.1 · THE DOME DOOR

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The hand bolt is intact and in the thrown (locked) position. The bolt keeper, the metal plate on the frame that receives the bolt, has been torn from the frame with its screws. The splintering of the frame is on the **inside face**. The pry marks from a flat bar are on the **inside edge of the door**. The outside face of the door and frame is undamaged.

Recorded as seen. No conclusion drawn by the original officer.

### E-1.2 · EXHIBITS RECOVERED

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| REF   | ITEM                                                                                | LOCATION FOUND                                                              |
|-------|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| EX-01 | Brass telescope counterweight, 34 mm handle with knurled grip, one of a set of four | Counterweight rack, platform. The other three are dusty; this one is clean. |
| EX-02 | Laptop, Prof. Hale, open, asleep, battery 9%                                        | Platform desk                                                               |
| EX-03 | Spectacles, Prof. Hale, one arm bent                                                | Floor, under the platform                                                   |
| EX-04 | Flat steel bar, 600 mm                                                              | Corridor outside the dome door                                              |
| EX-05 | Bolt keeper with two screws                                                         | Dome floor, inside the door                                                 |

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### E-1.3 · THE LADDER

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Vertical steel ladder, 2.4 m, rungs square-section with plain smooth edges. **No fitting on the ladder or platform rail carries a knurled or criss-cross surface.**

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## CASE REVIEW - HALE, EDMUND (DECEASED)

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On the morning of 7 January the director of the Ashcombe Observatory was found dead at the foot of a ladder, inside a telescope dome that had been bolted from the inside. The caretaker forced the door. The file was closed as an accidental fall.

His sister does not believe it. Neither, on paper, does the pathologist. The coroner has allowed a review.

### YOUR TASK

Decide whether Edmund Hale's death was an accident. If it was not, **name everyone responsible**, and show your working.

This is an Expert file. Not every document in it is what it says it is.

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### HOW THIS WORKS

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#### 1 · THE FILE OPENS IN THREE PHASES

Read what the memo names, write down your theory, then open the next bundle.

#### 2 · LOGS DON'T LIE. PEOPLE MIGHT. SO MIGHT A PRINTOUT.

If a statement and a log disagree, the log is right. **But a log is only as honest as the person who printed it.** Ask who supplied each record, and whether it is complete.

#### 3 · FIVE AUTHORITIES

Ten lines of inquiry are open. Solo and pairs get **five**; three or more players get six. Several of them return nothing you need.

#### 4 · WRITE IT DOWN AS YOU GO

Every step on the Case Resolution Sheet needs a citation. Your final finding may need more than one name.

#### 5 · ACCUSING

Two wrong accusations are allowed. Each costs 10 points and returns a *Reconsider* slip. The third is final. Then open **ENVELOPE S-1**.

## OPEN LINES OF INQUIRY

Ten authorities are available to this review. Each may be exercised once. Tick the box when spent and open the matching sealed slip.

| SPENT                    | NO. | AUTHORITY                                                                           | DIRECTED TO             |
|--------------------------|-----|-------------------------------------------------------------------------------------|-------------------------|
| <input type="checkbox"/> | 01  | Obtain the <b>cloud copy of the dome door log</b> direct from the lock manufacturer | KeyLine Systems         |
| <input type="checkbox"/> | 02  | Seize the <b>control-room workstation session log</b>                               | Digital Forensics       |
| <input type="checkbox"/> | 03  | Forensic examination of the <b>brass counterweight</b> , EX-01                      | Forensic Science        |
| <input type="checkbox"/> | 04  | Obtain <b>bank and telephone records - T. Brackley</b>                              | Bank / carrier          |
| <input type="checkbox"/> | 05  | Examine <b>Professor Hale's laptop</b> , EX-02, and his email drafts                | Digital Forensics       |
| <input type="checkbox"/> | 06  | Verify <b>Dr Okafor's travel</b> and hotel in Geneva                                | Airline / hotel         |
| <input type="checkbox"/> | 07  | Obtain the <b>doorbell camera</b> at Mrs Hale's home, Hallow Lane                   | Householder             |
| <input type="checkbox"/> | 08  | Obtain <b>bus CCTV and ticket data - K. Nowak</b>                                   | Valley Buses            |
| <input type="checkbox"/> | 09  | Commission a <b>second pathologist</b> to review the head injuries                  | Home Office pathologist |
| <input type="checkbox"/> | 10  | Obtain the <b>weather station and sky record</b> for the night                      | Observatory archive     |

## ADVICE FROM THE UNIT

Before you spend, ask what in the file made you want it, and whether you already hold the answer. Clearing a person you have already cleared is a luxury.

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**CASE RESOLUTION SHEET**

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Complete **before** making a final accusation. Cite evidence by document reference (e.g. B-1, D-1, SLIP 01). A step with no citation scores nothing.

**1. 1 · CAUSE - HOW DID HE DIE, AND WAS IT A FALL?**EVIDENCE **2. 2 · TIME - WHEN WAS HE LAST ALIVE, AND WHEN DID HE DIE?**EVIDENCE **3. 3 · THE LOCKED ROOM - WAS THE DOME REALLY BOLTED WHEN IT WAS FOUND?**EVIDENCE **4. 4 · THE FORGERY - WHICH RECORD CANNOT BE TRUSTED, AND HOW DO YOU KNOW?**EVIDENCE **5. 5 · THE KILLER - WHO WAS IN THE DOME WITH HIM, AND WHAT PROVES IT?**EVIDENCE **6. 6 · THE COVER-UP - WHO MADE IT LOOK LIKE AN ACCIDENT?**EVIDENCE

## FINDINGS ON EACH PERSON

Record each as Cleared, Suspect, Killer or Accomplice, with the evidence you rely on. A person is cleared only when a document puts them beyond the doubt.

## PERSON

## FINDING

## EVIDENCE RELIED ON

**BRACKLEY, Thomas**

Caretaker

**LINDQVIST, Dr Ivo**

Visiting researcher

**OKAFOR, Dr Miriam**

Deputy director

**HALE, Ellen**

Wife of the deceased

**NOWAK, Kasia**

Doctoral student

## MOTIVE (OPTIONAL - SCORES NOTHING)

## ACCUSATIONS

## NO.

## NAMED

## OUTCOME

1

2

3 -

## FINAL

## FINAL ACCUSATION

On the evidence recorded above, the death of Edmund Hale was:

☐ ACCIDENTAL      ☐ MURDER

and, if murder, it was committed by, and covered up by:

**NOW OPEN ENVELOPE S-1**